

BILKENT UNIVERSITY

		Engineering Mathematics Midterm		
Code : <i>MATH 241</i>		Last Name:		
Acad. Year: <i>2020-2021</i>		Name :		Student No.:
Semester : <i>Fall</i>		Department:		Signature:
Date : <i>14.11.2020</i>		4 QUESTIONS ON 4 PAGES TOTAL 100 POINTS		
Time : <i>8:40</i>				
Duration : <i>90 minutes</i>				
¹	²	³	⁴	

- No questions during the exam.
- This is a single exit exam, once you leave the classroom, you can not turn back.
- Please put your ID with a visible photo at a safe distance, for example, on the chair that is next to you.
- When the proctor comes to check your id, you must sign the attendance sheet again trying to keep the distance
- You can not leave before 9.00
- Once the exam is over, put your exam paper on the desk next to you. You will leave the room one by one starting from the person closest to the door. Once everyone leaves the room, we will collect your papers from your desk
- If you want to leave before 10.10, put your exam paper on the desk you are sitting and leave the room by keeping your distance with the proctor and with the other students.
- No partial credit will be given for unsupported answers!

1. (25 pts) Using Gauss-Jordan elimination, solve

$$x_1 + 3x_2 + x_4 = 2$$

$$x_1 + 4x_2 + 2x_3 = -1$$

$$-2x_2 - 2x_3 + x_4 = 0$$

$$2x_1 + 5x_2 + 2x_3 + x_4 = -5$$

$$x_1 + 2x_2 + 2x_3 = -7$$

Augmented matrix of the system is:

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 1 & 2 \\ 1 & 4 & 2 & 0 & -1 \\ 0 & -2 & -2 & 1 & 0 \\ 2 & 5 & 2 & 1 & -5 \\ 1 & 2 & 2 & 0 & -7 \end{array} \right] \xrightarrow{\substack{-R_1+R_2 \rightarrow R_2 \\ -2R_1+R_4 \rightarrow R_4 \\ -R_1+R_5 \rightarrow R_5}} \left[\begin{array}{cccc|c} 1 & 3 & 0 & 1 & 2 \\ 0 & 1 & 2 & -1 & -3 \\ 0 & -2 & -2 & 1 & 0 \\ 0 & -1 & 2 & -1 & -9 \\ 0 & -1 & 2 & -1 & -9 \end{array} \right] \xrightarrow{\substack{-3R_2+R_1 \rightarrow R_1 \\ 2R_2+R_3 \rightarrow R_3 \\ 3R_2+R_4 \rightarrow R_4 \\ R_2+R_5 \rightarrow R_5}} \left[\begin{array}{cccc|c} 1 & 0 & -6 & 4 & 11 \\ 0 & 1 & 2 & -1 & -3 \\ 0 & 0 & 2 & -1 & -6 \\ 0 & 0 & 4 & -2 & -12 \\ 0 & 0 & 4 & -2 & -12 \end{array} \right]$$

$$\xrightarrow{\substack{3R_3+R_1 \rightarrow R_1 \\ -R_3+R_2 \rightarrow R_2 \\ -2R_3+R_4 \rightarrow R_4 \\ -2R_3+R_5 \rightarrow R_5}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & -7 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & -1 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\frac{R_3}{2} \rightarrow R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & -7 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & -1/2 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + x_4 = -7$$

$$x_2 = 3$$

$$x_3 - \frac{x_4}{2} = -3$$

$$\left. \begin{array}{l} \text{let } x_4 = \alpha \Rightarrow x_3 = -3 + \frac{\alpha}{2} \\ x_2 = 3 \\ x_1 = -7 - \alpha \end{array} \right\} \Rightarrow$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -7 - \alpha \\ 3 \\ -3 + \frac{\alpha}{2} \\ \alpha \end{bmatrix}$$

$$= \begin{bmatrix} -7 \\ 3 \\ -3 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -1 \\ 0 \\ 1/2 \\ 1 \end{bmatrix}$$

is soln $\forall \alpha \in \mathbb{R}$.

2. (25 pts) Let $\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\{\mathbf{w}_1, \mathbf{w}_2\}$ be two linearly independent set of vectors in a vector space V .

If $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\} \cap \text{Span}\{\mathbf{w}_1, \mathbf{w}_2\} = \{\mathbf{0}\}$ then show that $\{\mathbf{u}_1 + \mathbf{w}_1, \mathbf{u}_2 + \mathbf{w}_2\}$ is also a linearly independent set.

want to show that if $\alpha_1(\underline{u}_1 + \underline{w}_1) + \alpha_2(\underline{u}_2 + \underline{w}_2) = \underline{0}$

then $\alpha_1 = \alpha_2 = 0$

since: $\alpha_1(\underline{u}_1 + \underline{w}_1) + \alpha_2(\underline{u}_2 + \underline{w}_2) = (\alpha_1 \underline{u}_1 + \alpha_2 \underline{u}_2) + (\alpha_1 \underline{w}_1 + \alpha_2 \underline{w}_2)$

if the sum is $\underline{0} \Rightarrow \alpha_1 \underline{u}_1 + \alpha_2 \underline{u}_2 = -(\alpha_1 \underline{w}_1 + \alpha_2 \underline{w}_2)$
 $\cap$
 $\text{Span}\{\underline{u}_1, \underline{u}_2\} \quad \text{Span}\{\underline{w}_1, \underline{w}_2\}$

$\Rightarrow \alpha_1 \underline{u}_1 + \alpha_2 \underline{u}_2 \in \underbrace{\text{Span}\{\underline{u}_1, \underline{u}_2\} \cap \text{Span}\{\underline{w}_1, \underline{w}_2\}}_{\substack{1) \\ \{\underline{0}\}}}$

$\Rightarrow \alpha_1 \underline{u}_1 + \alpha_2 \underline{u}_2 = \underline{0}$

$\Rightarrow \alpha_1 = \alpha_2 = 0$ since $\{\underline{u}_1, \underline{u}_2\}$ is a
L.I. set

Thus, $\{\underline{u}_1 + \underline{w}_1, \underline{u}_2 + \underline{w}_2\}$ is also a L.I. set.

3. (15+10 pts) Let V be the set of all real valued continuous functions defined on $[-1, 1]$ (i.e., $V = \{f(x) : f(x) \text{ is continuous in } [-1, 1]\}$). V is an inner space where the vector addition is

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x)$$

scalar multiplication is

$$(\alpha \mathbf{f})(x) = \alpha \mathbf{f}(x)$$

and the inner product is

$$\langle \mathbf{f}(x), \mathbf{g}(x) \rangle = \int_{-1}^1 \mathbf{f}(x) \mathbf{g}(x) dx.$$

The set S containing the polynomials of degree at most 2 is a subspace of V .

- Find an orthonormal basis for the subspace S .
- Find the orthogonal projection of $\mathbf{f}(x) = x^3$ on to S .

(a) Choose $\{1, x, x^2\}$ as a basis for $V = \{f: f(x) = a_0 + a_1x + a_2x^2, a_i \in \mathbb{R}\}$
 the set is L.I. because $\underbrace{0}_{\text{the zero function}} = a_0 + a_1x + a_2x^2 \Rightarrow a_0 = a_1 = a_2 = 0$

Then can obtain an orthonormal basis fns by using G-S process:

- $\underline{u}_1 = 1$, $\underline{q}_1 = \frac{\underline{u}_1}{\|\underline{u}_1\|}$, $\|\underline{u}_1\| = \sqrt{\int_{-1}^1 dx} = \sqrt{2} \Rightarrow \underline{q}_1 = \frac{1}{\sqrt{2}}$

• $\underline{u}_2 = x$, $\underline{e}_2 = \underline{u}_2 - \underline{q}_1 \langle \underline{q}_1, \underline{u}_2 \rangle = x - \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{2}} x' dx' = x$

$$\| \underline{e}_2 \| = \sqrt{\int_{-1}^1 x^2 dx} = \sqrt{\left[\frac{1}{3} x^3 \right]_{-1}^1} = \sqrt{\frac{2}{3}} \Rightarrow \underline{q}_2 = \sqrt{\frac{3}{2}} x$$

$$\bullet \underline{u_3} = x^2, \quad \underline{e_3} = \underline{u_3} - \underline{q_1} \langle \underline{q_1}, \underline{u_3} \rangle - \underline{q_2} \langle \underline{q_2}, \underline{u_3} \rangle = x^2 - \underbrace{\frac{1}{\sqrt{2}} \int_{-1}^1 \frac{1}{\sqrt{2}} x'^2 dx'}_{\substack{\text{odd func!} \\ 0}} - \underbrace{\sqrt{\frac{3}{2}} x \int_{-1}^1 \sqrt{\frac{3}{2}} x' x'^2 dx'}_0$$

$$\Rightarrow \underline{c_2} = x^2 - \left(\frac{\frac{1}{2} \cdot \frac{1}{3} x^3}{-1} \right) = x^2 - \frac{1}{3} ;$$

$$\|e_3\| = \sqrt{\int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx} = \sqrt{\frac{8}{45}} = \frac{2\sqrt{2}}{3\sqrt{5}} \Rightarrow \underline{a_3 = \frac{3\sqrt{5}}{2\sqrt{2}} \left(x^2 - \frac{1}{3}\right)}$$



(b) Projection of x^3 on $S = p(x) = \langle x^3, q_1 \rangle$

$$p(x) = \langle x^3, \underline{q_1} \rangle \underline{q_1} + \langle x^3, \underline{q_2} \rangle \underline{q_2} + \langle x^3, \underline{q_3} \rangle \underline{q_3}$$
$$= \frac{1}{\sqrt{2}} \underbrace{\int_{-1}^1 \frac{1}{\sqrt{2}} x^3 dx}_0 + \underbrace{\sqrt{\frac{3}{2}} x \int_{-1}^1 x'^3 \sqrt{\frac{3}{2}} x' dx'}_{\frac{\sqrt{6}}{5}} + \underbrace{\left(\frac{3\sqrt{5}}{2\sqrt{2}}\right) \left(x^2 - \frac{1}{3}\right) \int_{-1}^1 \left(x'^2 - \frac{1}{3}\right) x'^3 dx'}_0$$

$$\Rightarrow \underline{\underline{p(x) = \frac{3}{5} x}}$$

4. (10+15 pts)

(a) Given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -5$, find $\begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix}$

(b) Let $A = \begin{bmatrix} a & b & 0 & 0 & 0 \\ c & d & 0 & 0 & 0 \\ e & f & 0 & 0 & 0 \\ g & h & i & j & k \\ l & m & n & o & p \end{bmatrix}$. How many different values can you obtain for $\det(A)$?

Explain your reasoning.

a) Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \xrightarrow{-3R_1 \rightarrow R_1} \begin{bmatrix} -3a & -3b & -3c \\ d & e & f \\ g & h & i \end{bmatrix} \xrightarrow{-4R_2 + R_1, -4R_3 + R_1} \begin{bmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{bmatrix}$

" B

$$\det B = -3 \det A = +15$$

b) Consider $(a,b), (c,d), (e,f) \in \mathbb{R}^2$. Since $\mathbb{R}^2$ is 2 dim, any set of 3 vectors in $\mathbb{R}^2$ is linearly dependent i.e., $\exists \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ s.t. $\alpha_1(a,b) + \alpha_2(c,d) + \alpha_3(e,f) = (0,0)$ has nontrivial solns. This implies that $\alpha_1(a,b,0,0,0) + \alpha_2(c,d,0,0,0) + \alpha_3(e,f,0,0,0) = (0,0,0,0,0)$ has nontrivial solns. That is, one of the rows of A , is a linear combination of others. Hence, doesn't matter what the entries of A are, its determinant takes just one value which is zero.